



PROJECT QUALITY MANAGEMENT PLAN TEMPLATE

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QUALITY MANAGEMENT PLAN DOST STARBOOKS: WHIZ CHALLENGE

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APRIL 2026





TABLE OF CONTENTS

INTRODUCTION	3
QUALITY MANAGEMENT APPROACH	3
QUALITY REQUIREMENTS / STANDARDS	5
QUALITY ASSURANCE	6
QUALITY CONTROL	8





INTRODUCTION

The Quality Management Plan is a core component of the overall project management plan for the DOST STARBOOKS Whiz Challenge. This plan establishes how quality will be defined, planned, assured, and controlled throughout the full lifecycle of the project — from initiation through final deployment and handover to DOST-STII.

The Quality Management Plan for the STARBOOKS Whiz Challenge project will guide Group NEXUS in delivering a gamified, offline-first educational platform that meets the expectations of DOST-STII and the end users it serves. The purpose of this plan is to:

- Ensure that quality is deliberately planned into every phase of the project and not treated as an afterthought
- Define how quality will be managed across both the product and the development processes
- Establish quality assurance activities that verify process compliance throughout the project lifecycle
- Establish quality control activities that verify the product meets defined standards at key checkpoints
- Define measurable quality standards that serve as the basis for acceptance

QUALITY MANAGEMENT APPROACH

The quality management approach for the STARBOOKS Whiz Challenge will ensure that both product quality and process quality are addressed in an integrated and continuous manner. Quality will be built into the project from the outset by defining clear standards, measuring outcomes against those standards, and continuously improving based on findings. This approach applies across all five project phases: Initiation, Planning, Executing, Monitoring & Controlling, and Closing.

Product quality for the STARBOOKS Whiz Challenge is defined by the functional and non-functional requirements established in the Software Requirements Specification (SRS) and agreed upon with DOST-STII. The focus is on delivering a working application — covering all four game modes (Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle), the Player Module, and the Admin Dashboard — that meets the product acceptance criteria defined in the Scope Management Plan, performs reliably on existing DOST kiosk hardware, and satisfies DOST-STII's standards for educational content and platform integrity.

Process quality for the STARBOOKS Whiz Challenge will focus on the development, testing, and documentation practices used across all sprints. Given that this project follows an Agile/Scrum methodology, process quality checkpoints are embedded into the sprint cycle. Establishing process quality standards ensures that development activities are consistent, predictable, and aligned with the project's scope baseline and schedule commitments.



The Project Manager, in coordination with the QA/Test Lead, will work with the entire project team to identify, define, and document all relevant quality standards for both product and processes. These will be incorporated into the project documentation maintained in OpenProject and communicated to all stakeholders — including DOST-STII representatives and the Faculty Adviser — at sprint reviews and through the Communications Management Plan's reporting channels.

Metrics will be established and tracked throughout the project lifecycle to measure quality at both the product and process levels. The QA/Test Lead is responsible for working with the team to define these metrics, conduct measurements at defined intervals, and surface results for review. Product and process measurements will be reviewed by the Project Sponsor (DOST-STII) at each sprint review as one criterion in assessing the project's progress and delivery readiness. Metrics will include:

- **Schedule:** Sprint velocity and task completion rate versus planned sprint commitments
- **Resources:** Actual hours logged per team member per sprint versus planned hours (10 hours/week/member)
- **Cost:** Actual cost per phase versus phase budget allocations in OpenProject
- **Process performance:**
 - Sprint task completion rate (percentage of sprint backlog items completed within the sprint)
 - Defect discovery rate per sprint (number of bugs found during internal testing)
 - Defect resolution time (average time from bug report to resolution)
- **Product performance:**
 - Feature completeness (percentage of acceptance criteria passed per sprint deliverable)
 - System performance on kiosk hardware (load time, offline functionality, stability)
 - Pilot testing feedback scores (usability, engagement, and educational value ratings from student users)
- **Customer satisfaction:** Sprint review feedback from DOST-STII stakeholders on each delivered increment

Quality improvements may be identified by any member of the project team, the Faculty Adviser, or DOST-STII during sprint reviews or retrospectives. Each recommendation will be evaluated for its impact on scope, schedule, and deliverable quality. If an improvement is approved, the Project Manager will update all relevant project documents in OpenProject, and the change will be communicated to all stakeholders in accordance with the Communications Management Plan.



QUALITY REQUIREMENTS / STANDARDS

This section describes the quality requirements and standards that govern both the product and processes of the STARBOOKS Whiz Challenge. These standards form the baseline against which all deliverables and development activities will be measured.

Product Quality:

Product quality standards for the STARBOOKS Whiz Challenge are derived from the product acceptance criteria defined in the Scope Management Plan, the functional and non-functional requirements in the SRS, and the direct input of DOST-STII during sprint reviews. These standards govern whether the delivered system is considered acceptable.

The project team will know that the product complies with quality standards when all of the following are consistently demonstrated:

- All four game modes operate on DOST kiosk hardware without crashes, scoring errors, or gameplay defects across repeated test runs.
- The system functions fully offline for all core player-facing features, with Whiz Battle and AI question generation activating only when connectivity is confirmed.
- The Admin Dashboard enables non-technical DOST-STII staff to perform all administrative functions — content management, player management, analytics viewing, and reward processing — independently and without errors.
- The application runs within acceptable performance thresholds on existing STARBOOKS kiosk hardware, with no degradation under normal usage conditions.
- Pilot testing with student participants produces consistently positive ratings for usability, engagement, and educational value, with no critical usability issues reported.
- The final deliverable contains no critical bugs, data integrity failures, or unresolved system crashes.

Any product quality standard that cannot be met using existing organizational norms will be identified by the project team and escalated to the Faculty Adviser and DOST-STII for validation before being incorporated into the project documentation.

Process Quality:

Process quality standards for the STARBOOKS Whiz Challenge are based on the Agile/Scrum practices adopted by Group NEXUS and the sprint-level commitments tracked in OpenProject. These standards ensure that the development process itself is disciplined, traceable, and continuously improving.

Process quality compliance will be assessed by the Project Manager and QA/Test Lead at the close of each sprint. The project will be considered process-compliant when:



- Sprint task completion rates consistently meet or exceed the team's established sprint velocity baseline.
- Defects discovered during quality control activities are logged, prioritized, and resolved within the same or immediately following sprint.
- All code submissions include associated test cases and documentation consistent with the SRS and SDD standards.
- Sprint reviews with DOST-STII confirm alignment between delivered features and the agreed sprint backlog prior to each review.
- All project documentation is kept current in OpenProject following each sprint, with scope, cost, and schedule records accurately reflecting completed work.

QUALITY ASSURANCE

Quality assurance for the STARBOOKS Whiz Challenge focuses on the processes used to develop, test, and deliver the platform across all sprints. An iterative quality assurance process is embedded into the project's Scrum lifecycle, ensuring that process compliance is monitored continuously rather than evaluated only at the end of the project.

The Project Manager and QA/Test Lead will conduct process assessments at defined intervals — specifically at the end of each sprint — to verify that all development and documentation processes are being properly executed. Key performance metrics for process quality assurance include sprint task completion rate, defect discovery rate, defect resolution time, and documentation completeness. The established project tolerances for these metrics are based on the team's sprint velocity baseline and the phase budget and schedule commitments in OpenProject.

The table below provides the key quality assurance metrics for the STARBOOKS Whiz Challenge project:

Process Action	Acceptable Process Standards	Process Phase	Assessment Interval
Sprint Execution	≥ 80% of sprint backlog items completed within the sprint; velocity deviation ≤ 20% from baseline	Executing	End of each sprint
Defect Management	All critical and high-priority defects resolved within the same or immediately following sprint; defect resolution	Executing / Monitoring & Controlling	Per sprint and on demand



	time $\leq$ 5 business days for critical issues		
Code Submission	100% of submitted code accompanied by associated test cases; all submissions reviewed by at least one other team member before merge	Executing	Per feature / pull request
Documentation Completeness	All sprint deliverables documented in OpenProject within 2 business days of sprint close; SRS, SDD, and management plan updates reflected within the same sprint	Executing / Planning	End of each sprint
Sprint Review Readiness	100% of features presented at sprint review have passed internal QA testing and meet defined acceptance criteria prior to the review session	Executing	Prior to each sprint review

The QA/Test Lead will conduct process reviews on a per-sprint basis, monitor process performance metrics in OpenProject, and ensure all development activities comply with project and organizational standards. Findings, including any identified discrepancies or non-conformances, will be documented and brought to the attention of the Project Manager for review.

The Project Manager will incorporate a review of process compliance, quality assurance findings, and process improvement initiatives as a standing agenda item in all sprint retrospectives and scheduled management reviews.

Process improvement is a continuous responsibility. All quality assurance reviews, retrospective outcomes, and assessment findings should produce actionable improvements to the team's development or documentation practices. All approved improvements will be documented, incorporated into the relevant project plans in OpenProject, and communicated to all stakeholders.



QUALITY CONTROL

Quality control for the STARBOOKS Whiz Challenge focuses primarily on the application itself — monitoring and verifying that all delivered features and system components meet the product quality standards defined in the Scope Management Plan and SRS. Quality control activities are performed at designated checkpoints throughout the Executing phase and culminate in comprehensive system testing prior to final deployment.

All features developed during each sprint will be submitted to the QA/Test Lead for internal testing prior to the sprint review with DOST-STII. Additionally, integrated system testing will be conducted at the completion of all core features before the pilot testing phase begins. Physical deployment and performance testing on DOST kiosk hardware will be conducted during the final preparation phase before the June 2026 launch.

The table below defines the product quality standards and assessment activities for the STARBOOKS Whiz Challenge:

Product / Feature	Performance / Quality Standards	Quality Assessment Activities	Assessment Intervals
All Four Game Modes (Whiz Challenge, Memory Match, Puzzle, Battle)	Zero critical bugs or game-breaking defects; accurate scoring and badge award logic; stable gameplay across repeated sessions on kiosk hardware	Functional testing per feature using defined test cases from the SRS; regression testing after each sprint	Per sprint and prior to pilot testing
Player Module (Registration, Login, Profile, Leaderboard, Rewards)	All player data stored and retrieved accurately; session/token management functioning correctly; leaderboard rankings accurately reflect game results	Integration testing between player-facing features and backend API; user acceptance testing during pilot	Per sprint (feature-level) and pre-launch (integration)
Admin Dashboard	All management functions (player management, quiz content, reward processing, analytics)	End-to-end functional testing of all admin workflows; usability review with	Prior to pilot testing and sprint review



	operate without errors; accessible to non-technical users without external support	DOST-STII representative	
Offline Functionality	All core features function fully without internet access; Whiz Battle and AI question generation correctly activate only when connectivity is available	Offline simulation testing on kiosk hardware; connectivity toggle testing	Pre-launch and during kiosk deployment testing
Kiosk Performance	Application loads within acceptable time thresholds; no performance degradation under normal multi-user conditions on existing STARBOOKS hardware	Performance testing on target DOST kiosk hardware; load testing for concurrent usage scenarios	Prior to pilot deployment and final launch
Pilot Testing (Student Users)	Positive usability, engagement, and educational value ratings; no critical usability issues reported; all feedback documented	Structured pilot testing session with student participants; post-session feedback collection	Once, prior to final deployment

The project team will conduct all functional and integration testing using test cases defined in the project's Testing and Documentation Package. Performance and deployment testing on actual kiosk hardware will be coordinated with DOST-STII. DOST-STII representatives will participate in sprint reviews as the formal acceptance checkpoint for each increment and will provide sign-off on the final deliverable at project closure.

The Project Manager will include a standing agenda item in all sprint reviews and project status meetings for the review of product quality findings, any open defects, and product improvement initiatives.



Meeting all defined product quality standards is essential to the success of the STARBOOKS Whiz Challenge. Doing so ensures that the platform delivers meaningful educational value to student users at DOST events, enables DOST-STII staff to manage the system independently, and results in a deployment that is stable, reliable, and ready for long-term operational use.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:



Marievic V. Narquita
Information Services Section (ISS) Head of DOST-STII

Date: April 8, 2026

